higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING (CAD COMPONENT)

(Second Paper)

NQF LEVEL 3

4 MARCH 2019

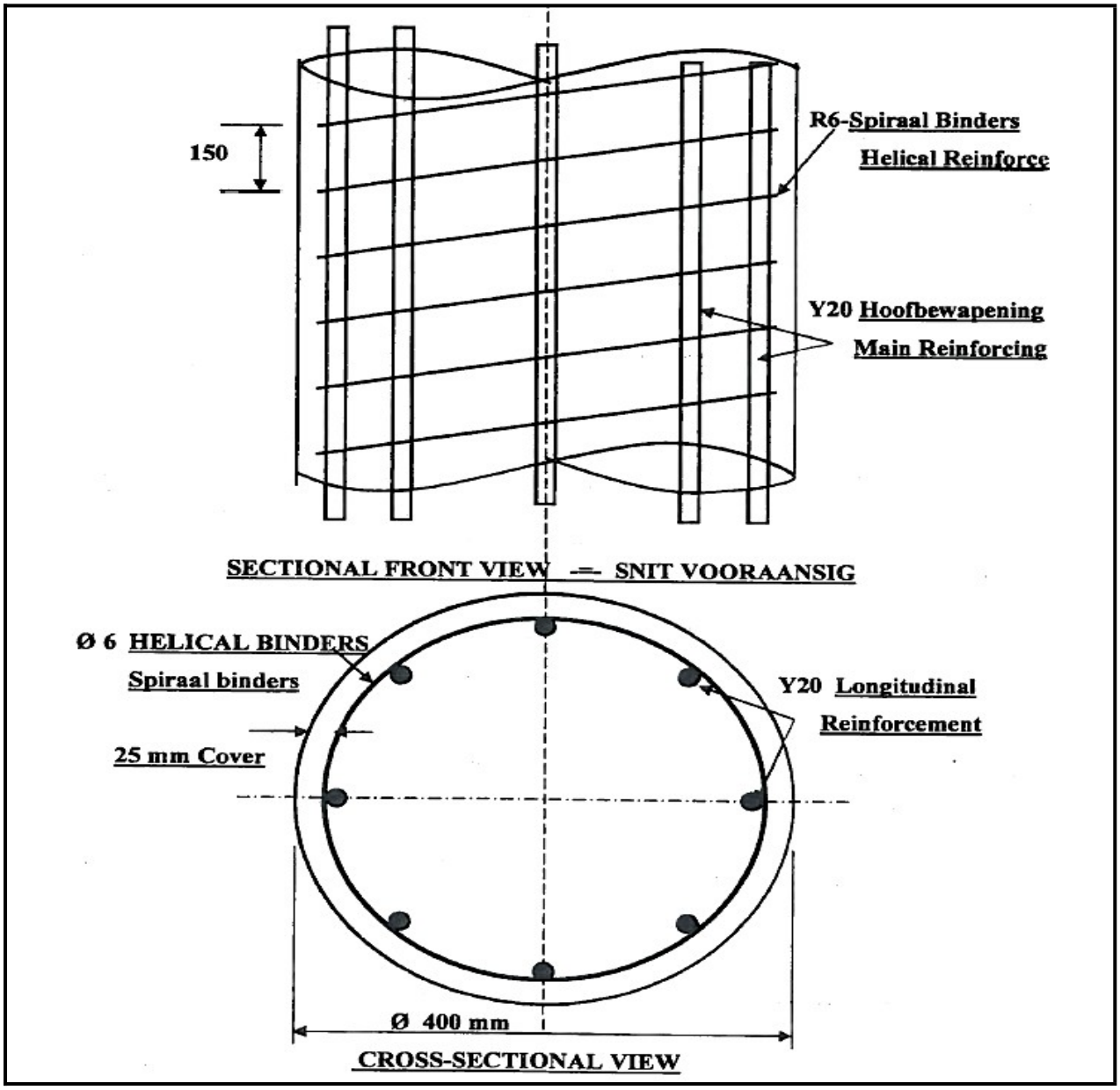
This marking guideline consists of 6 pages.

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NOTE TO MARKERS

This marking guideline is NOT absolute and serves as a guideline only. Consider alternative correct answers.

QUESTION 1: REINFORCED-CONCRETE DETAILING



CROSS-SECTIONAL VIEW MARK ALLOCATION	
Correct scale	2
Correctness of detail	4
Labelling	2
Line work	2
TOTAL:	**10**

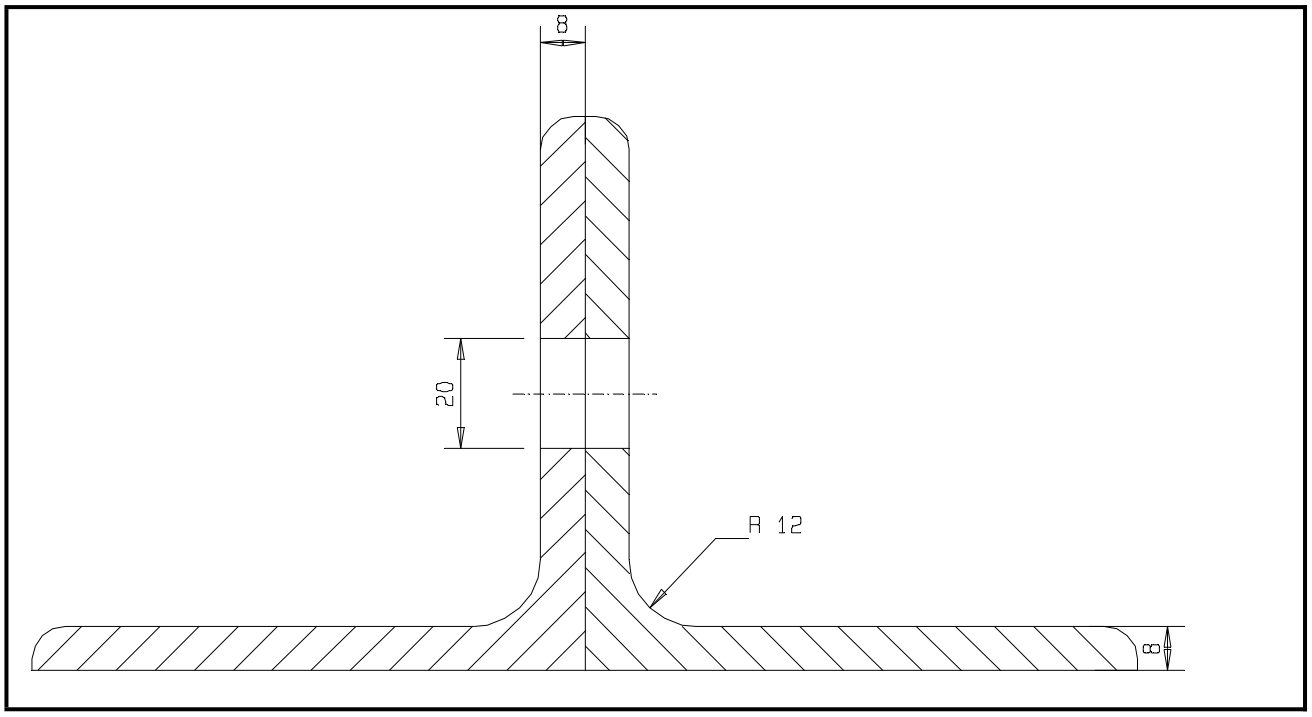
(10)

FRONT-SECTIONAL VIEW MARK ALLOCATION	
Correct scale	2
Correctness of detail	5
Labelling	2
Line work	1
TOTAL:	**10**

(10)
[20]

QUESTION 2: STRUCTURAL STEEL WORK DETAILING

2.1

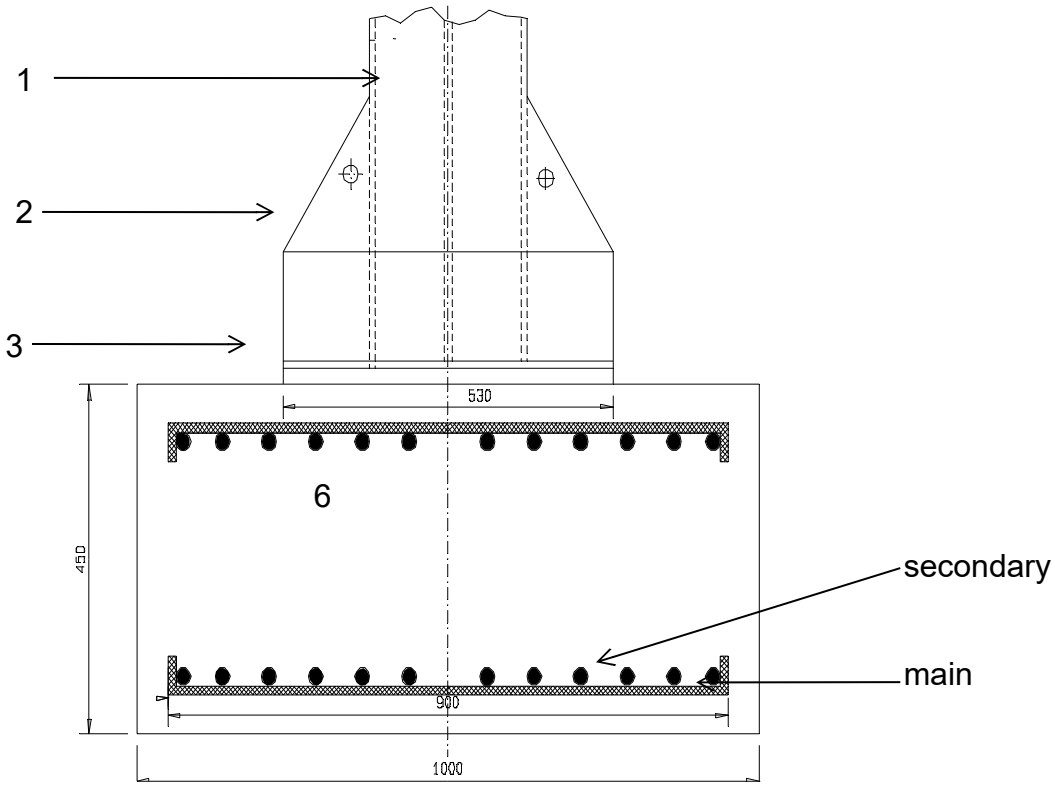


RSA 100 × 100 × 8
SCALE 1:1

Correctness	4
Dimensions	2
Line work	1
Correct scale	2
Title and scale	1
TOTAL:	**10**

(10)

2.2



NOTES

ITEM	MEMBER	SIZES
1	H-beam	254 × 254 × 79,9 kg per metre
2	Gusset plate	All bolt holes are drilled on the centre lines of the angles and all bolts are M24.Gusset has two equal sides and parallel but perpendicular to the base side.
3	Unequal angle beam	175 × 75 × 10,2 off
4	Base plate	530 mm × 530 mm × 20 mm
5	Concrete cover	50 mm
6	Concrete reinforcement	Compression and tension bars: 2Y12 Stirrup rods: 12R10

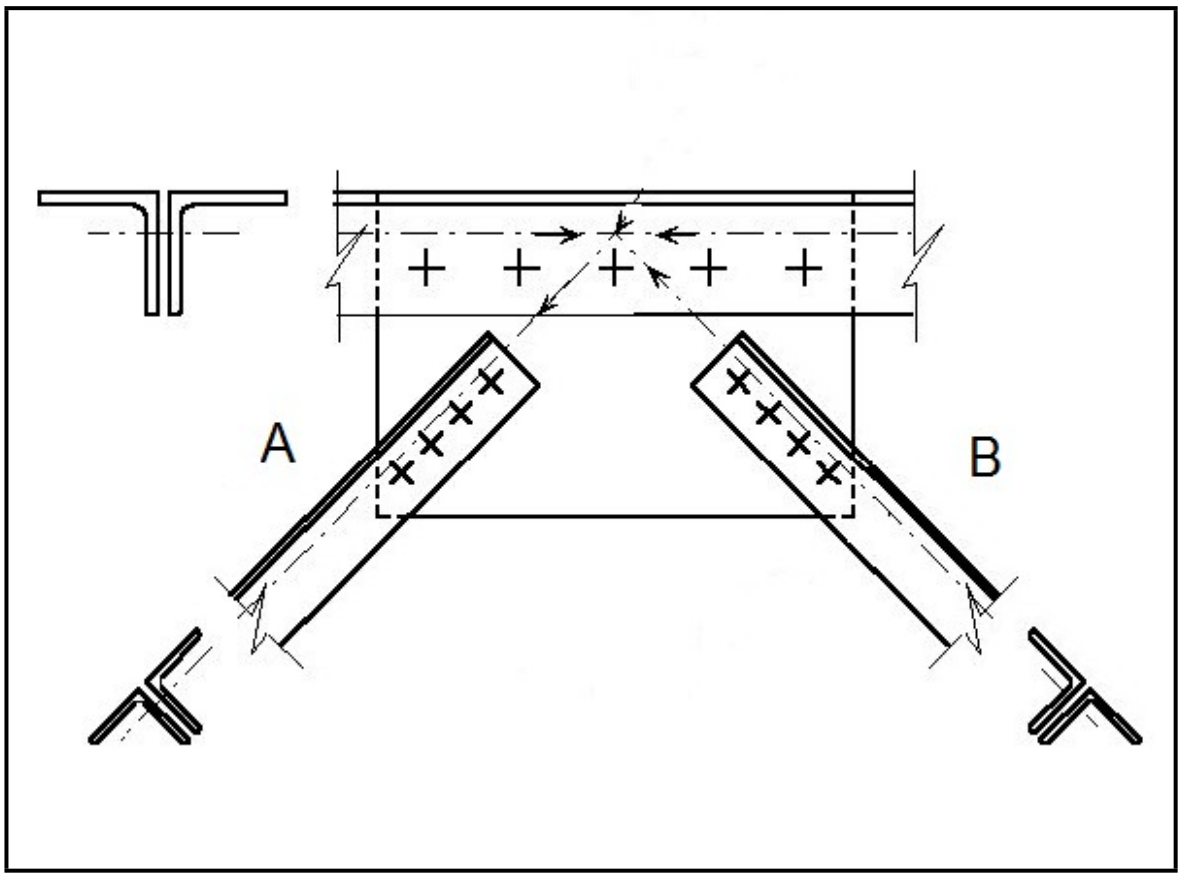
MARK ALLOCATION	
Correct scale	2
Correct drawing detail (angle plate with bolt holes, base plate column and foundation)	6
Reinforcement	5
Notes	4
Essential dimensions	3
TOTAL:	**20**

Consider any suitable answers. (20)

[30]

QUESTION 3: STRUCTURAL STEEL WORK DETAILING

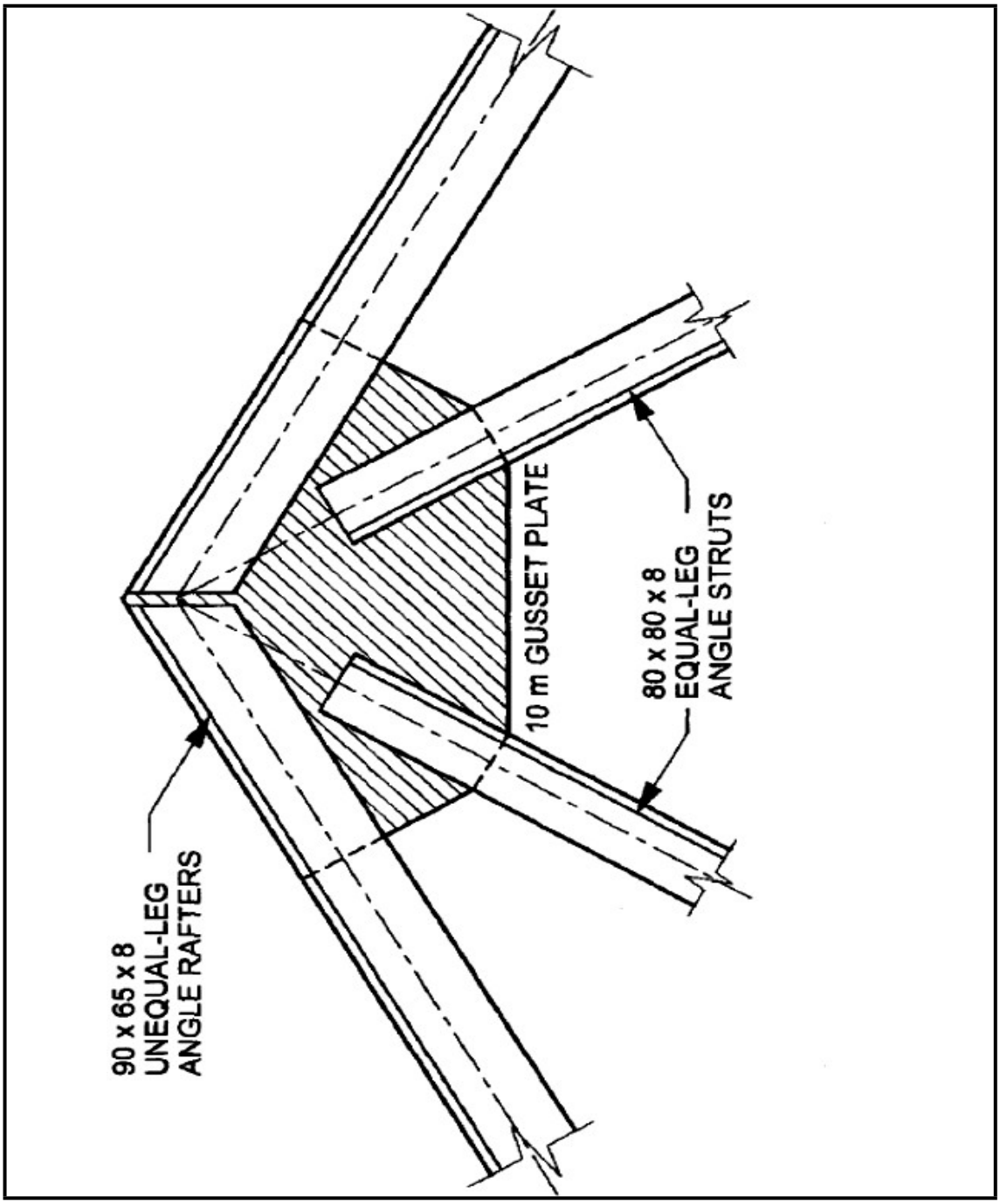
3.1



MARK ALLOCATION	
Correct scale	2
Top-chord detail correct	5
Gusset plate	3
Line work	2
Rolled steel detail shown	3
TOTAL:	**15**

(15)

3.2



MARK ALLOCATION	
Setting out correct	1
Two rafters correct	3
Two struts correct	2
Gusset plate	2
Line work	2
TOTAL:	**10**

(10)
[25]

TOTAL: 75

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